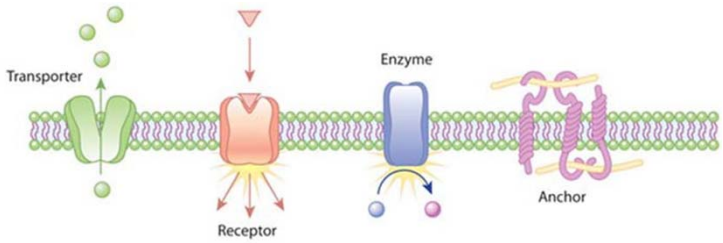
Protein Structure-Function relationship

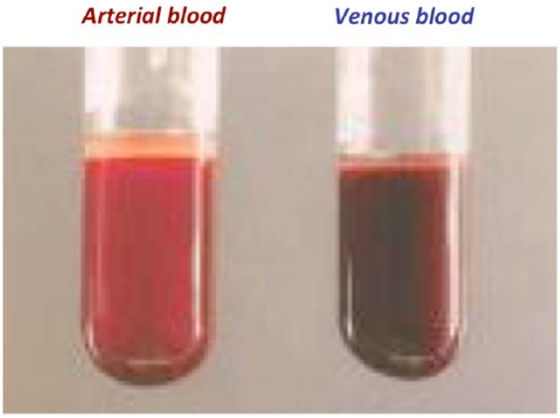


Protein Function

Learning goals:

- Why can't plasma oxygen support the body's oxygen needs?
- Why blood hemoglobin solves the problem of oxygen supply?
- How hemoglobin achieves two different affinities for oxygen
- What's an allosteric regulator and how it works
- What are the allosteric regulators for hemoglobin and how they work
- What's the mechanism of CO poisoning

Blood and Oxygen Transport



Pulse Oximeters Measure Oxygen Saturation



Calculates the oxy/deoxy hemoglobin ratio; converts to percent saturation

This image shows a sample of arterial blood (left) and venous blood (right). The difference in color is a result of the fact that the hemoglobin molecule has a different pattern of light absorption when it is bound to oxygen. Clinically, this difference can be measured with a simple device called the pulse oximeter.

The Central Problem: Limited Oxygen Solubility in Water (Plasma)

- Our bodies consume about 250 mL of oxygen per minute at rest
- Heart pumps 5 liters of blood/minute (cardiac output)
- Only 3 mL of oxygen can dissolve in one liter of plasma
- Thus only 15 mL of dissolved oxygen can be delivered per minute by our cardiovascular system (6% of the total needed at rest!)
- Solution: Hemoglobin, the major protein contained in red blood cells (RBCs)

Determining the Hematocrit

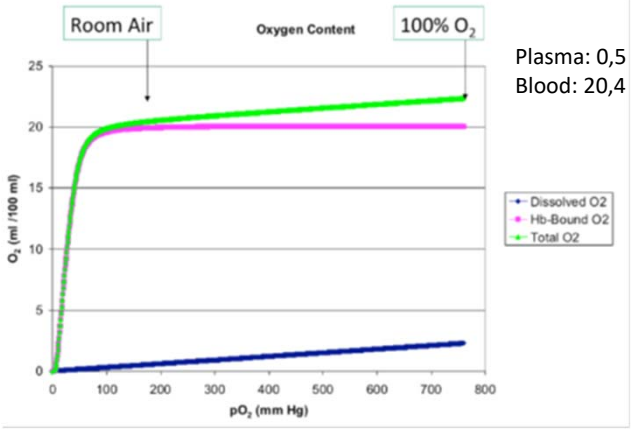
1) Draw blood
2) Add anticoagulant
3) Centrifuge

Plasma

Volume of packed red cells:
Hematocrit



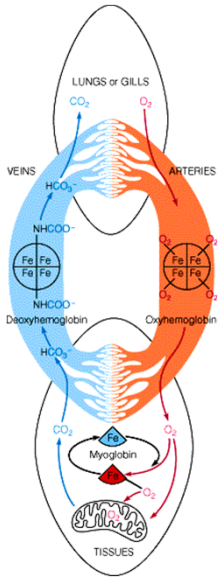
Blood vs Plasma Oxygen content



Plasma: 0,5 ml O_2/100 ml
Blood: 20,4 ml O_2/100 ml

Blood transports 4 times more oxygen that we need at rest¡

Role of the globins in oxygen transport and storage



Globins Are Oxygen-Binding Proteins

Biological problems:

•Protein side chains lack affinity for O_2.

•Some transition metals bind O_2 well but would generate free radicals if free in solution.

•Organometallic compounds such as heme are more suitable, but Fe^{2+} in free heme could be oxidized to Fe^{3+} (very reactive!).

Biological solution:

- Capture the oxygen molecule with heme that is protein bound.

Myoglobin (storage) and hemoglobin (transport) can bind oxygen via a protein-bound heme.

Structures of Porphyrin and Heme

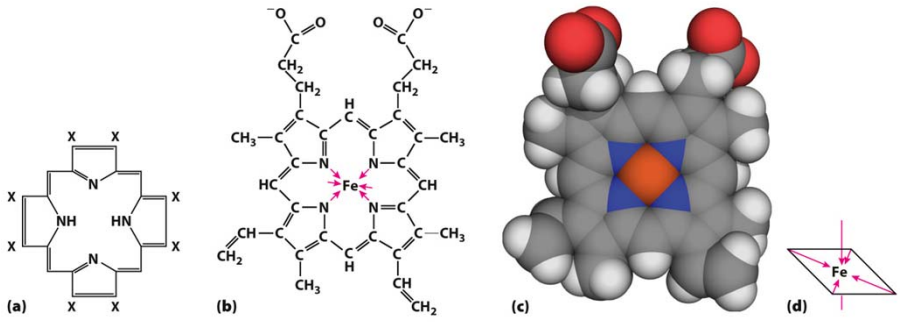


Figure 5-1

Protoporphyrin ring | Heme group | Heme group

Structure of Myoglobin

The heme group viewed from the side

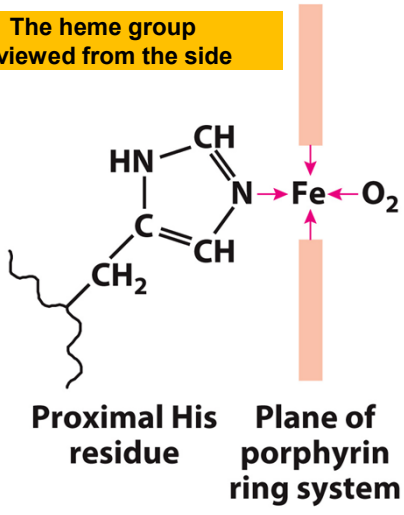


Figure 5-2
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Myoglobin tertiary structure

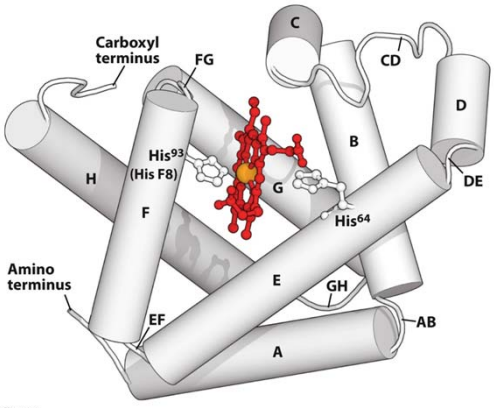


Figure 5-3
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Could Myoglobin Transport O_2?

- pO_2 in lungs is about 13 kPa: it sure binds oxygen well.
- pO_2 in tissues is about 4 kPa: it will not release it!

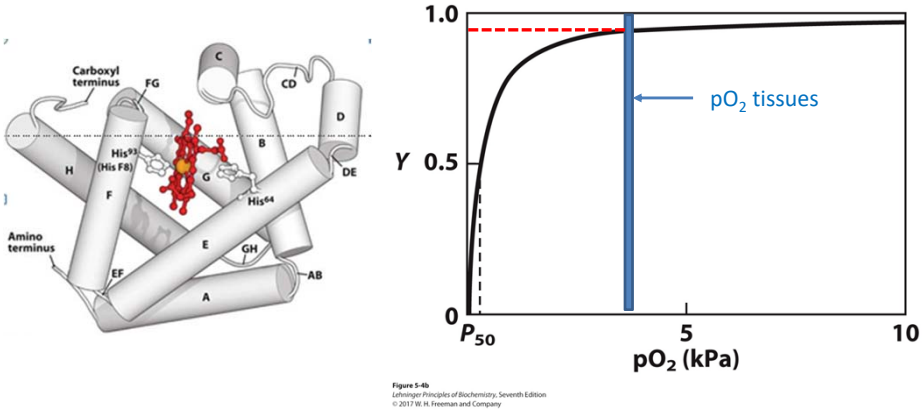


Figure 5-4b
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Would lowering the affinity (P_{50}) of myoglobin to oxygen help?

For Effective Transport Affinity Must Vary with pO_2

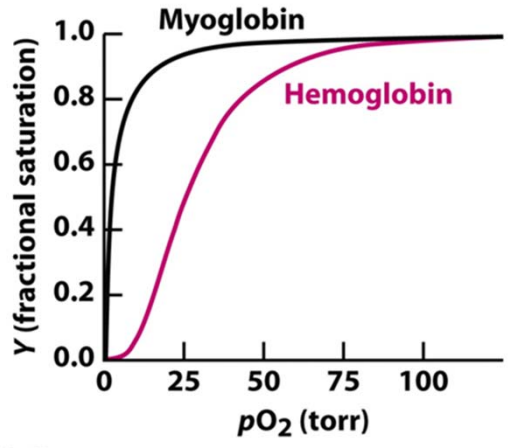
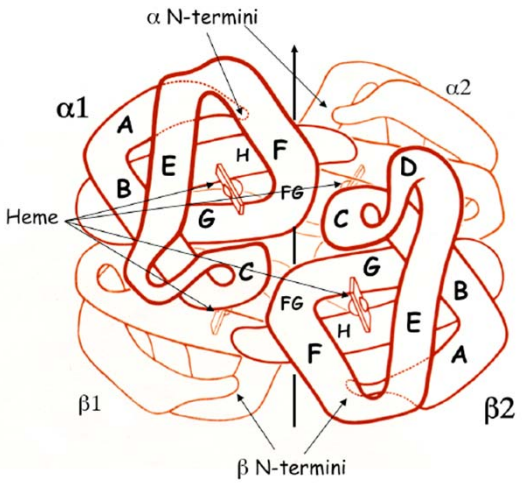


Figure 7-7
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How Can Affinity to Oxygen Change?

- It must be a protein with multiple binding sites.
- Binding sites must be able to interact with each other.
- This phenomenon is called cooperativity.
  - positive cooperativity
    - first binding event increases affinity at remaining sites
    - recognized by sigmoidal binding curves
  - negative cooperativity
    - first binding event reduces affinity at remaining sites

Structure of Hemoglobin



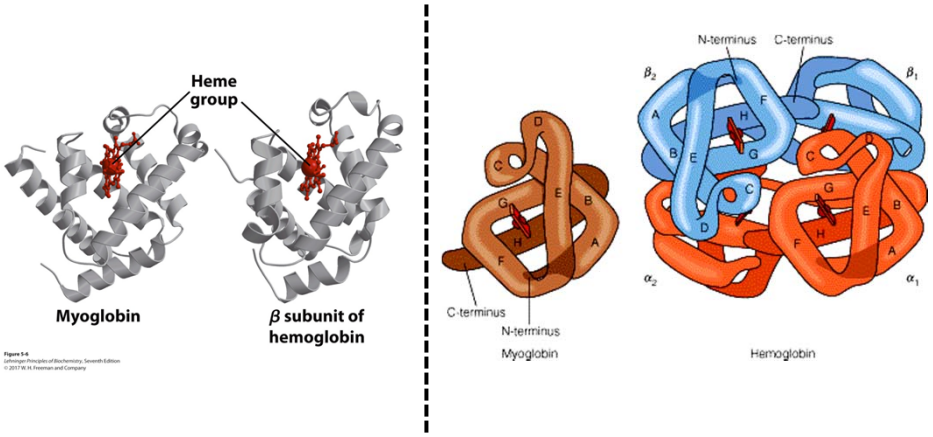
- Hb is an $\alpha_2\beta_2$ tetramer.
- Each subunit has a myoglobin-like tertiary structure with 8 α-helices that fold around a heme "prosthetic" group.
- The heme groups are covalently tethered to each subunit by an Fe-histidine linkage.
- The α chains have 141 residues.
- The β chains have 146 residues.

Cooperativity Is a Special Case of Allosteric Regulation

- Allosteric protein
  - Binding of a ligand to one site affects the binding properties of a different site on the same protein.
  - can be positive or negative
  - homotropic
    - The normal ligand of the protein is the allosteric regulator.
  - heterotropic
    - A different ligand affects binding of the normal ligand.
- Cooperativity = positive homotropic regulation

Hemoglobin Binds Oxygen Cooperatively

- Hemoglobin (Hb) is a tetramer of two subunits ($\alpha 2 \beta 2$).
- Each subunit is similar to myoglobin.



Subunit Interactions in Hemoglobin

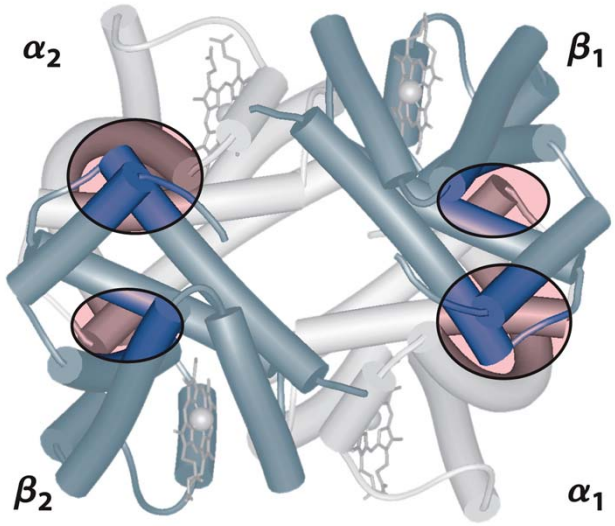
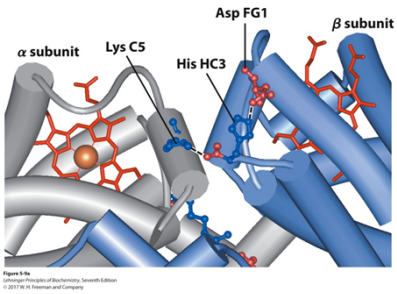


Figure 5-8
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Subunit Interactions: Details



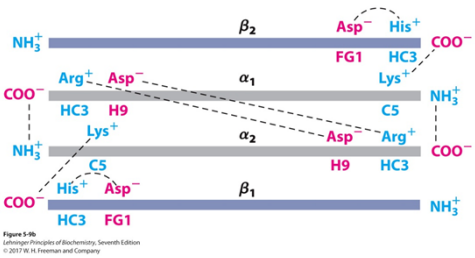
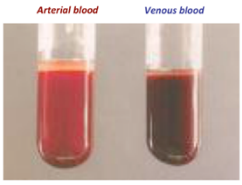


Figure 5-9b
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Some ion pairs that stabilize the T state of deoxyhemoglobin

Spectroscopic Detection of Oxygen Binding to Globins

- The heme group is a strong chromophore that absorbs both in ultraviolet and visible range.
- Ferrous form (Fe^{2+}) without oxygen has an intense Soret band at 429 nm.
- Oxygen binding alters the electronic properties of the heme and shifts the position of the Soret band to 414 nm.
- Binding of oxygen can be monitored by UV-Vis spectrophotometry.
- Deoxyhemoglobin (in venous blood) appears purplish in color and oxyhemoglobin (in arterial blood) is red.



Pulse Oximeters Measure Oxygen Saturation



Calculates the oxy/deoxy hemoglobin ratio; converts to percent saturation

R and T States of Hemoglobin

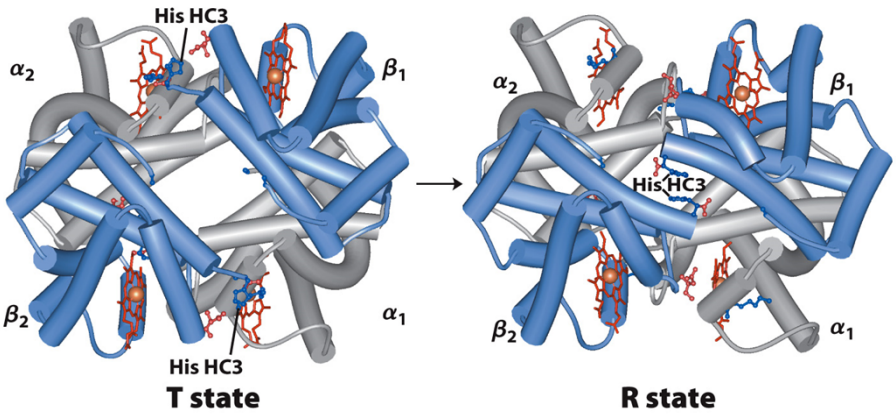
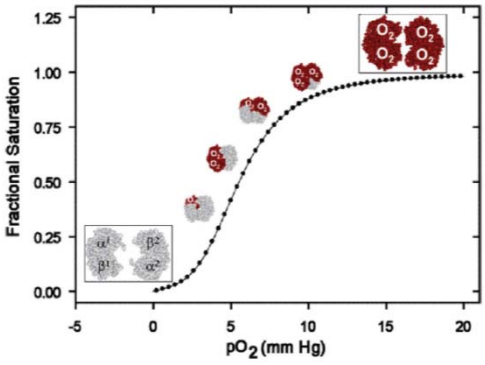


Figure 5-10
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O2 binding triggers a T → R conformational change.

- T = tense state
  - more interactions, more stable
  - lower affinity for O_2
- R = relaxed state
  - fewer Interactions, more flexible
  - higher affinity for O_2
- Conformational change from the T state to the R state involves breaking ion pairs between the α1-β2 interface.



Ackers & Holt. JBC 2006

Conformational Change Is Triggered by Oxygen Binding

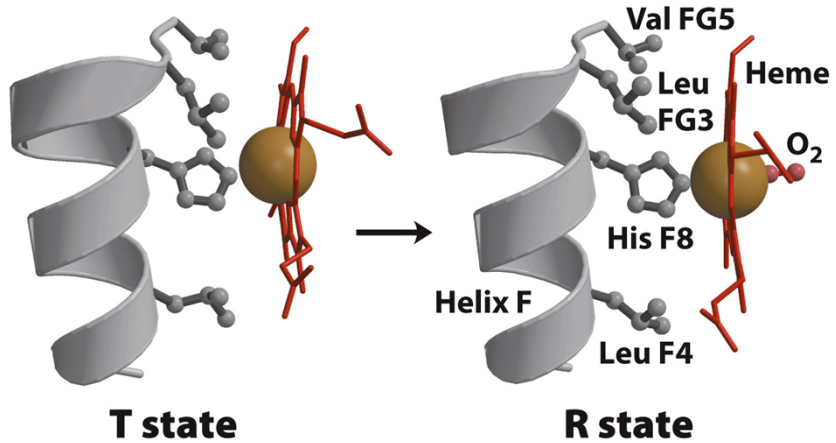


Figure 5-11
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Subunit Interactions: Details

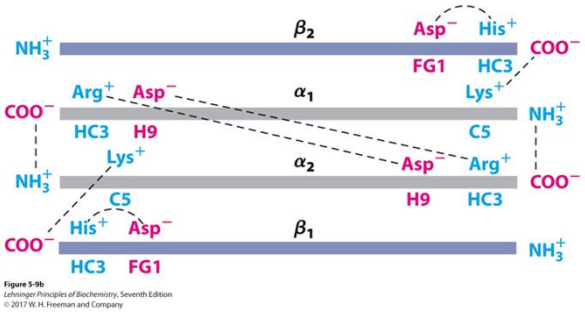


Figure 5-9b
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Some ion pairs that stabilize the T state of deoxyhemoglobin must be broken during the oxygen binding

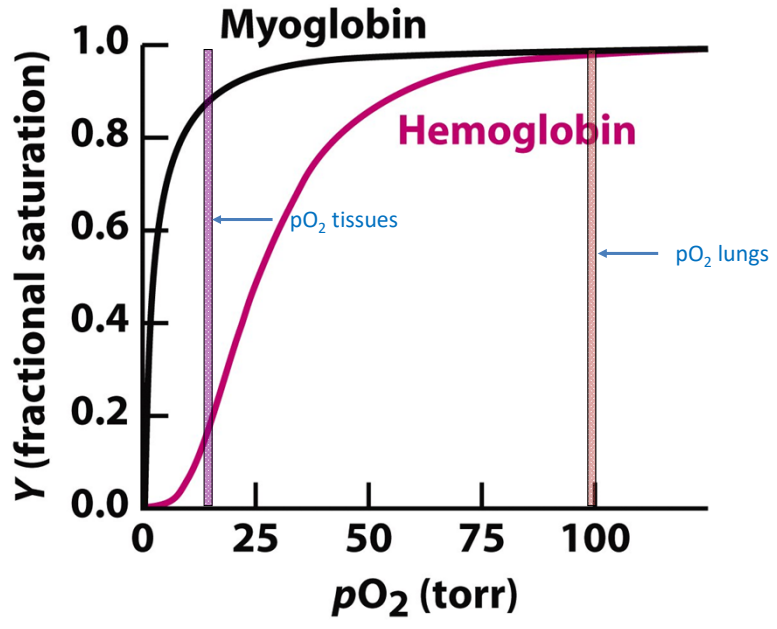


Figure 7-7
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Physiological factors that regulate oxygen affinity to Hemoglobin

The Bohr effect: influence of H^+ and CO_2 concentration on O_2 affinity to Hemoglobin



pH Effect on O_2 Binding to Hemoglobin

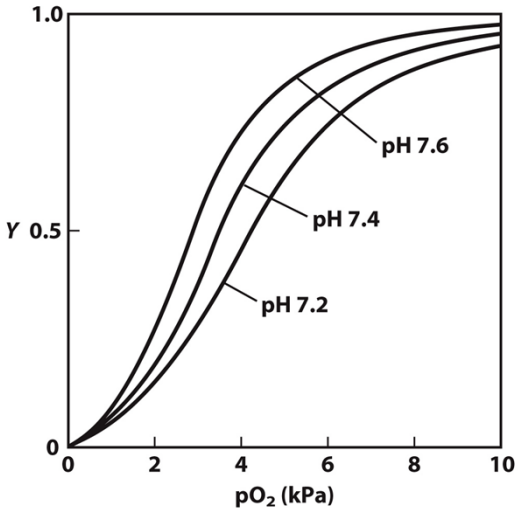


Figure 5-16
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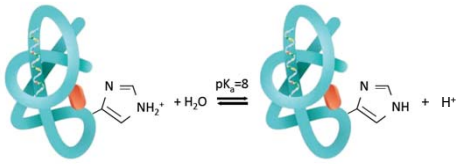
Does acidity increase or decrease the K_d?

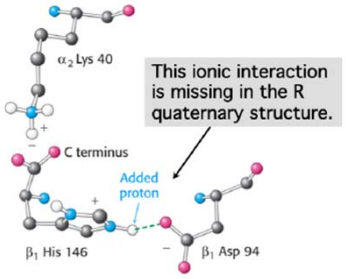
pH Effect on O_2 Binding to Hemoglobin

- Actively metabolizing tissues generate H^+, lowering the pH of the blood near the tissues relative to the lungs (catalyzed by carbonic anhydrase).

$$CO_2 + H_2O \leftrightarrow H_2CO_3 \leftrightarrow HCO_3^- + H^+$$

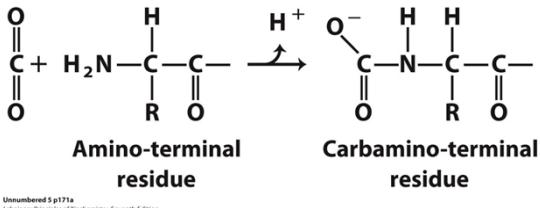
- Hb Affinity for oxygen depends on the pH.
  - H^+ binds to Hb and stabilizes the T state.
    - protonates His146, which then forms a salt bridge with Asp94
    - leads to the release of O_2 (in the tissues)
- The pH difference between lungs and metabolic tissues increases efficiency of the O_2 transport.
- This is known as the Bohr effect.





Hemoglobin and CO_2 Export

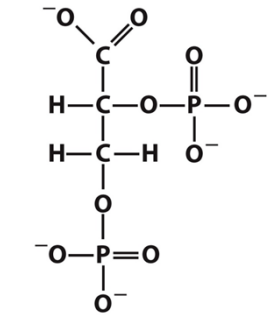
- CO_2 is produced by metabolism in tissues and must be exported.
- 15–20% of CO_2 is exported in the form of a carbamate on the amino terminal residues of each of the polypeptide subunits.



- Notice:
  - The formation of a carbamate yields a proton that can contribute to the Bohr effect.
  - The carbamate forms additional salt bridges, stabilizing the T state.

2,3-Bisphosphoglycerate Regulates O_2 Binding

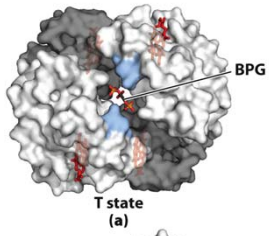
- Negative heterotropic regulator of Hb function
- Present at mM concentrations in erythrocytes
  – produced from an intermediate in glycolysis
- Small negatively charged molecule, binds to the positively charged central cavity of Hb
- Stabilizes the T states

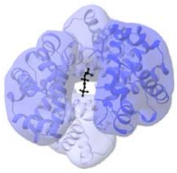


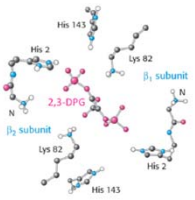
2,3-Bisphosphoglycerate

Unnumbered 5 p171b
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2,3-BPG Binds to the Central Cavity of hB







DPG (2,3-diphosphoglycerate) binds preferentially to the quaternary T structure and lowers the O_2 affinity of a solution of hemoglobin by shifting the T-R equilibrium toward the low affinity T structure.

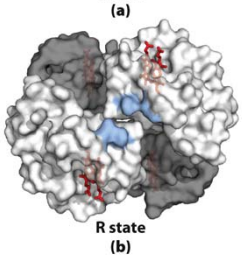


Figure 5-18
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2,3-BPG Allows for O_2 Release in the Tissues and Adaptation to Changes in Altitude

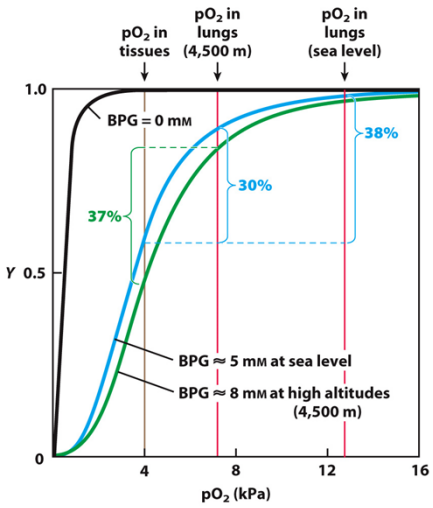
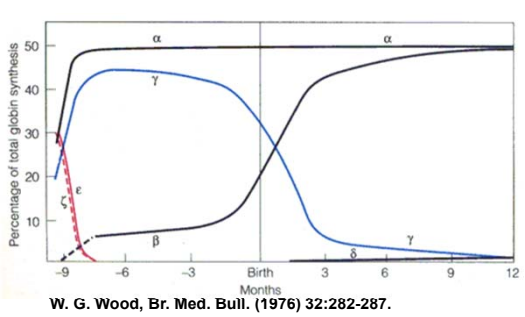


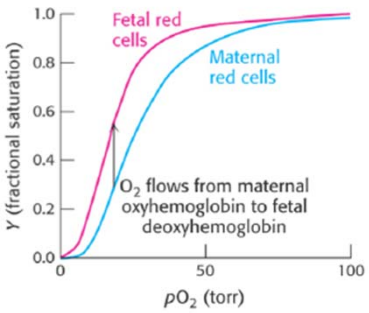
Figure 5-17
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Fetal Hemoglobin quaternary structure binds 2,3-BPG with less affinity thus binds O_2 more efficiently than maternal

Expression of human globin genes at different stages of development.



W. G. Wood, Br. Med. Bull. (1976) 32:282-287.



Caso clínico

Parámetro	Rango de referencia
pH arterial	7.35 - 7.45
pCO_2 arterial	32 - 48 mmHg
pO_2 arterial	83 - 108 mmHg
CO_2 total arterial	23 - 29 mEq/l
HCO_3^- arterial	20-28 mEq/l
Exceso de base arterial	Entre –2 y +3 mEq/l
Saturación de O_2 arterial	95 - 98 %

Un niño de 15 años fue encontrado inconsciente en su casa. Según la declaración de sus padres, el chico se encontraba en su habitación, que está junto a la habitación donde se encuentra la caldera de gas que calienta el agua sanitaria y de calefacción. Fue trasladado a un hospital, donde se encontró que estaba comatoso con incontinencia de orina y cianosis. Los gases en sangre arterial mostraron pH 7,3, $PaCO_2$ 31,8, PaO_2 40,9 mmHg, HCO_3^- 16 meq/L, SaO_2 72% y SBE -8,5. Se encontró hipoxemia. El TAC cerebral reveló edema, y la radiografía de tórax mostró un edema pulmonar bilateral. El nivel de carboxihemoglobina arterial (COHb) una hora después de ser encontrado inconsciente fue del 51,9 %. Se le intubó y se suministró O_2 al 100%. Cuatro horas después del ingreso, la COHb arterial era del 8,4%. A las 10-12 horas del ingreso, el cuadro pulmonar mejoró progresivamente. La presión arterial también se volvió más estable. Durante los días siguientes, el estado cardiovascular y respiratorio del paciente siguió mejorando. Fue extubado con éxito el quinto día. Los hallazgos neurológicos fueron normales después de la extubación. Fue trasladado de la unidad de cuidados intensivos pediátricos a la planta de ingreso pediátrico donde ya no requirió oxígeno.

Adaptado from Chang-Teng et al. Cases Journal 2009

Binding of Carbon Monoxide

- CO has similar size and shape to O_2; it can fit to the same binding site.
- CO binds heme over 20,000 times better than O_2 because the carbon in CO has a filled lone electron pair that can be donated to vacant *d*-orbitals on the Fe^{2+}.
- The protein pocket decreases affinity for CO, but it still binds about 250 times better than oxygen.
- CO is highly toxic, as it competes with oxygen. It blocks the function of myoglobin, hemoglobin, and mitochondrial cytochromes that are involved in oxidative phosphorylation.

CO vs. O_2 Binding to Free Heme

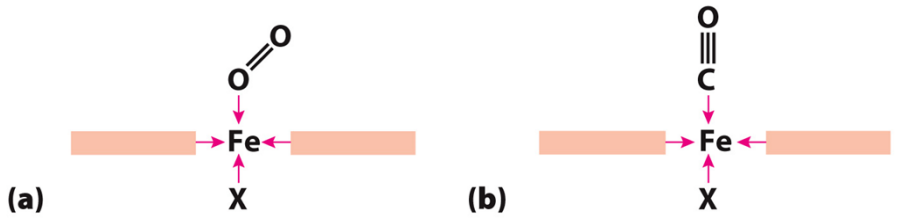


Figure 5-5ab
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Heme Binding to Protein Affects CO vs. O_2 Binding

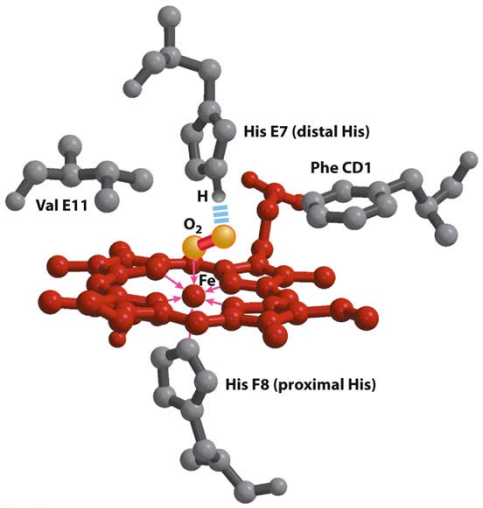
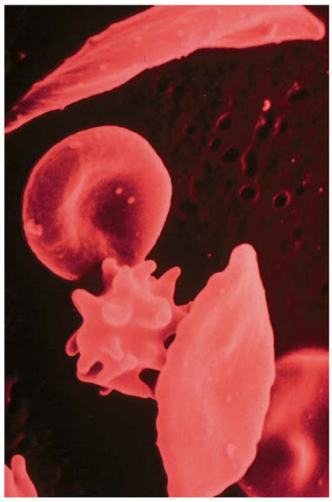


Figure 5-5c
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Sickle-Cell Anemia Is Due to a Mutation in Hemoglobin

- Glu6 → Val in the β chain of Hb
- The new Valine side chain can bind to a different Hb molecule to form a strand similar to the amyloidgenic proteins discussed previously.
- This sickles the red blood cells.
- Untreated homozygous individuals generally die in childhood.
- Heterozygous individuals exhibit a resistance to malaria.



Jackie Lewin, Royal Free Hospital/Science Source

Figure 5-19b
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Formation of Hb Strands in Sickle-Cell Anemia

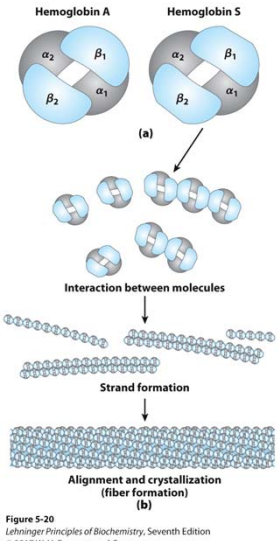


Figure 5-20
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